

Ontology metadata recommendations by TIB

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Table of contents

0 Executive summary

1 Why Ontology Metadata?

- 1.1 How to provide ontology metadata
- 1.2 Why ontology metadata?

2 How to read this document

- 2.1 Chapter structure and wording
- 2.2 Metadata recommendations and SHACL
- 2.3 Prefixes used in this document
- 2.4 Acronyms

3 Ontology-level metadata

- 3.1 Mandatory Metadata
 - 3.1.1 Ontology title
 - 3.1.2 Preferred ontology prefix/ acronym
 - 3.1.3 Ontology license
 - 3.1.4 Ontology creator(s)
 - 3.1.5 Ontology version IRI
 - 3.1.6 Ontology creation date
 - 3.1.7 Ontology abstract
 - 3.1.8 Ontology issue tracker
 - 3.1.9 Ontology documentation
- 3.2 Recommended Metadata
 - 3.2.1 Ontology contributor(s)
 - 3.2.2 Ontology funder
 - 3.2.3 Ontology funding
 - 3.2.4 Ontology audience description
 - 3.2.5 Ontology subject(s)
 - 3.2.6 Ontology annotation language(s)
 - 3.2.7 Applied logical framework
 - 3.2.8 Ontology serialization/ file format
 - 3.2.9 Ontology status
 - 3.2.10 Ontology code repository
 - 3.2.11 Ontology distributions/ products
 - 3.2.12 Application example
 - 3.2.13 Related resource(s)
 - 3.2.14 Citation suggestion
 - 3.2.15 Ontology sources (derived from)
 - 3.2.16 Ontology root classes
- 3.3 Optional Metadata
 - 3.3.1 Ontology description
 - 3.3.2 Alternative ontology title
 - 3.3.3 Alternative ontology prefix/ acronym
 - 3.3.4 Related version/ version history
 - 3.3.4.1 Prior ontology version
 - 3.3.4.2 Related ontology version
 - 3.3.4.3 Compatible ontology version
 - 3.3.4.4 Incompatible ontology version
 - 3.3.5 Social media
 - 3.3.6 KOS type
 - 3.3.7 Example ontology identifier
 - 3.3.8 Ontology identifier pattern
 - 3.3.9 Ontology homepage
 - 3.3.10 Ontology publisher

- 3.3.11 Ontology comments
- 3.3.12 Example ontology class
- 3.3.13 Ontology mailing list
- 3.3.14 Ontology logo/ depictions/ related visualization
- 3.3.15 Alternative identifiers of the ontology
- 3.3.16 Development environment
- 3.3.17 Alignments/ mappings
 - 3.3.17.1 Aligned resources
 - 3.3.17.2 Alignment files/ mapping files
- 3.3.18 Competency questions
- 3.3.19 Applied methodology
- 3.3.20 Preferred ontology namespace
- 3.3.21 Ontology issue date
- 3.3.22 Modification date
- 3.3.23 Textual version information
- 3.3.24 Version notes
- 3.4 Tabular overview - recommendations
- 3.5 Relations to related work

4 Term-level Metadata

- 4.1 Mandatory metadata
- 4.2 Recommended metadata
- 4.3 Optional metadata
- 4.4 Tabular overview - recommendations
- 4.5 Relations to related work

5 Sources

0. Executive summary

The number of ontology registries, archives and catalogues has increased over time. There are multiple services that list and also index ontologies. The registry entries of such services usually require the metadata of the ontologies they index, in order to make the ontologies findable and to provide users with general introductory information about the ontologies and their reuse. Registries currently face several problems, though:

- It is not yet standard procedure to provide rich ontology metadata despite efforts towards increasing the FAIRness of semantic resources [1], [2], in particular legacy ontologies may not be updated with rich metadata.
- Metadata may be provided in unstructured form outside of the ontology source which makes their collection difficult.
- Different registries may provide structured, open metadata documents for the same ontology which may at some point become incongruous with the source data.
- Currently, no standard metadata schema for ontologies exists.

In order to avoid these problems, metadata management should be part of the ontology engineering process: The metadata of an ontology should be part of the ontology's code base or the ontology itself. This would make the ontology resp. its code base the single source of truth for its own metadata.

It is therefore mandatory that an ontology provides its own metadata with machine-understandable semantics and established terms from metadata ontologies and controlled vocabularies. The metadata need to be as or even more persistent than the ontology itself. [Principle R1](#) of the FAIR principles [1] furthermore recommends to not just provide metadata that allow the discovery of an ontology (e.g. title, ontology prefix, authors), but to add metadata that richly describe the contents of the ontology and the conditions of its creation and use (cf. [1.2 Why ontology metadata?](#)).

With this document, *TIB - Leibniz Information Centre for Science and Technology University Library* provides a practical guide for metadata management of ontologies. It gives recommendations on required, recommended and optional metadata for ontologies. As there are currently several recommendations available (cf. [13], [14], [15], [16], [17], [18], [19], [20], [21], [22], [23]), this guide will provide recommendations on ontology metadata that particularly adress the publishing of ontologies on the [TIB Terminology Service](#).

In addition, we provide shapes for SHACL validators based on these recommendations (cf. [2.2 Metadata recommendations and SHACL](#)).

1 Introduction

1.1 How to provide ontology metadata?

Ontology metadata must be provided in a machine-understandable format. We therefore recommend to provide them as a part of the ontology itself: The metadata should be statements about the ontology that use dedicated annotation, object or data properties, to guarantee a good level of granularity and distinguishability of the metadata.

If you provide metadata that are here considered as *recommended* or *optional* in a single file, you may add a statement using `rdfs:seeAlso` to point to that document. Ideally, this document should then be etrievable with a persistent identifier (PID) - e.g. a DOI or a PURL. However, we do not recommend this, especially if that document is not machine-understandable.

1.2 Why ontology metadata?

Providing rich, machine-understandable metadata for an ontology has many advantages for their users (cf. [1], [2]), e.g.:

- They give a rich description about the ontology including its provenance, scope and purpose.
- They allow automatic extraction and indexing of the ontology by different services and reduce the effort of manual curation.
- They enable others to reuse the ontology or its terms and apply them for their own purposes.
- They indicate the current status of an ontology, distinguishing whether an ontology is still maintained by an active community or whether it is dormant.
- They allow referencing other Web resources with persistent identifiers (e.g. resources with identifiers like [DOI](#), [ROR](#), [ORCID](#), [RAiD](#), etc.).
- They allow detailed requests about ontologies on the web.
- They enable comparability and automated comparisons of ontologies.
- They can be retrieved via APIs and used in many different services.

2 How to read this document

2.1 Chapter structure and wording

This guide tries to use clear wording to distinguish absolute requirements (*must*, *required*, *mandatory*) from either recommendations (*should*, *recommended*) or optional metadata (*may*, *can*, *could*, *optional*). The sub-sections will each give

- a short introduction to the metadatum
- recommend a property
- recommend controlled vocabularies where appropriate
- provide examples in text/turtle serialization
- list alternative properties that may also be used to provide the metadatum
- specify SHACL validation rules (cf. section [2.2 Metadata recommendations and SHACL](#))

Subproperties to any properties mentioned will not be implied, but explicitly listed in the "alternative properties" part of each section.

2.2 Metadata recommendations and SHACL

This guide is a human-readable specification. In addition, we also provide SHACL specifications of these recommendations that can be/are used for three different purposes.

1. Data validation

A shape for data validation can be retrieved from <https://www.purl.org/ontology/metadata/shape> and can be used with SHACL validators to test ontologies for any violations of required metadata by [TIB Terminology Service](#) and to get suggestions for recommended metadata. We recommend the online SHACL validators [SHACL Playground](#), [SHACL Playground by Zazuko](#) or [SHACL Play!](#) for performing evaluations. Turtle or json-ld are common input format. SHACL Play! also allows to use IRIs for shapes and ontologies.

The shape can also be used to evaluate metadata of instances of `skos:ConceptScheme`. The instances of `skos:ConceptScheme` need to be declared as instances of `owl:Ontology` to trigger the shapes (cf. also section [4.6.2. SKOS Concept Schemes and OWL Ontologies](#) of the [SKOS Reference document](#)). A video demonstration of how to use this is given at our GitHub repository: <https://github.com/user-attachments/assets/c5d6be07-3bfb-44ab-ae65-e3f75b8e883a>.

2. Data validation at [TIB Terminology Service](#)

The shape at <https://www.purl.org/ontology/metadata/shape4ts> is applied in the Ontology Suggestion Feature at [TIB Terminology Service](#). The messages are adapted to the context of use and the particular use case: The person suggesting an ontology is not necessarily involved in its development. On TIB Terminology Service, they will be asked to provide metadata of the ontology if these are not present in the ontology, and if known to the user. These users are not required to provide the metadata as code. A video demonstration on how this is applied is given at our GitHub repository: <https://github.com/user-attachments/assets/847cfd39-0823-4a83-9c8a-0cf3d23d38a7>

3. Data generation

A shape for form generators and code generation can be retrieved from <https://www.purl.org/ontology/metadata/shape4forms>. It can for example be used with the [form generator tool by ULB Darmstadt](#) which has a [live demo instance](#). This shape is not as granular as the one for validation: All constraints are bundled in one shape in order to generate a form, that users can use to enter values. The user input is validated immediately for all criteria and the metadata code is generated. Only the recommended properties will be used to do so: This version of the recommendation does not make use of `sh:alternativePath` constructs, in order to generate valid RDF code with named properties, not anonymous/ blank nodes. A video demonstration on how to apply this is given at our GitHub repository: <https://github.com/user-attachments/assets/55cdf44f-5289-4989-9014-5e670fd73418>.

All three versions of the shape have the target class `owl:Ontology`, so that they can only be used to validate entities that declare themselves as an instance of `owl:Ontology`, and also code generated with these shapes, will declare entities as instances of `owl:Ontology`.

2.3 Prefixes used in this document

Code examples will usually provide full IRIs of statement subjects and objects. Properties will be presented in prefixed notation. The table provides the prefix definitions:

prefix	namespace
adms	http://www.w3.org/ns/adms#
bibo	http://purl.org/ontology/bibo/

prefix	namespace
dc	http://purl.org/dc/elements/1.1/
dcat	http://www.w3.org/ns/dcat#
dcterms	http://purl.org/dc/terms/
doap	http://usefulinc.com/ns/doap#
foaf	http://xmlns.com/foaf/0.1/
idot	http://identifiers.org/idot/
mod	https://w3id.org/mod#
obo	http://purl.obolibrary.org/obo/
omv	http://omv.ontoware.org/2005/05/ontology#
owl	http://www.w3.org/2002/07/owl#
pav	http://purl.org/pav/
premis	http://www.loc.gov/premis/rdf/v3/
rdf	http://www.w3.org/1999/02/22-rdf-syntax-ns#
rdfs	http://www.w3.org/2000/01/rdf-schema#
sdo	https://schema.org/
sh	http://www.w3.org/ns/shacl#
vann	http://purl.org/vocab/vann/
void	http://rdfs.org/ns/void#
xsd	http://www.w3.org/2001/XMLSchema#

2.4 Acronyms

Acronym	Full form
DOI	Digital Object Identifier
IRI	International Resource Identifier
PID	Persistent Identifier
ROR	Research Organization Identifier

3 Ontology-level Metadata

3.1 Mandatory metadata

Metadata in this section are mandatory for publishing an ontology at [TIB Terminology Service](#). Ontologies missing these metadata will not be accepted.

3.1.1 Ontology title

Ontologies must state their own title. The title will be displayed on [TIB Terminology Service](#), e.g. on each [ontology's landing page](#), in [search results](#) or in [ontology lists](#). The ontology must have at most one title per language. Ontology titles should be tagged for language.

Recommended property: <http://purl.org/dc/terms/title>

Example (text/turtle):

```
<https://www.purl.org/SomeOntology> rdf:type owl:Ontology ;
    dcterms:title "Some Ontology"@en .
```

Alternative properties:

- <http://purl.org/dc/elements/1.1/title>
- <https://schema.org/name>
- <https://schema.org/headline>

SHACL validation rules:

- `sh:datatype rdf:langString`
- `sh:minCount 1`
- `sh:uniqueLang true`

3.1.2 Preferred ontology prefix/ acronym

The ontology must declare its preferred, unique prefix or a short acronym. The prefix/ acronym will be used on TIB Terminology Service's [ontology list](#) as a short name for the ontology. It must not contain hyphens or other special characters and should be written in lowercase.

You can check whether an ontology prefix is already in use on services like [prefix.cc](#) or [Bioregistries](#).

Recommended property: <http://purl.org/vocab/vann/preferredNamespacePrefix>

Example (text/turtle):

```
<https://www.purl.org/SomeOntology> rdf:type owl:Ontology ;
    vann:preferredNamespacePrefix "so"^^xsd:string .
```

Alternative properties:

- <http://identifiers.org/idot/preferredPrefix>
- <https://w3id.org/mod#acronym>

SHACL validation rules:

- `sh:datatype xsd:string`

3.1.3 Ontology license

Ontologies must declare their license, referring to their license document via PID. The license text helps others to evaluate how they may reuse the ontology. Only ontologies with an open license will be published on [TIB Terminology Service](#). Consider, for example, the [list of licenses](#) that are conformant with the [Open Definition](#).

Recommended property: <http://purl.org/dc/terms/license>

Example (text/turtle):

```
<https://www.purl.org/SomeOntology> rdf:type owl:Ontology ;
    dcterms:license <http://creativecommons.org/licenses/by/4.0/> .
```

Alternative properties:

- <https://schema.org/license>
- <http://creativecommons.org/ns#license>
- <http://dbpedia.org/ontology/license>
- <http://purl.org/dc/terms/licence> (is a mis-spelled variant of <http://purl.org/dc/terms/license>)

SHACL validation rules:

- value must be from defined list of accepted licenses
- value must not be from defined list of unaccepted licences
- `sh:maxCount 1`
- `sh:minCount 1`

3.1.4 Ontology creator(s)

The ontology must list its creators, i.e. the people or institutions who were responsible for its development. It is recommended to refer to a creator with a PID (e.g. [ORCID](#), [Wikidata](#)-ID, or [ROR](#)-ID). Plain name strings can be provided in addition for readability. If an ontology is developed by a larger group, it is recommended to give the organization or project identifier as the creator and list individual persons as contributors. If no project or organisation PID is available, the respective name (`rdf:langString`) will also be accepted.

Recommended property: <http://purl.org/dc/terms/creator>

Example 1 (text/turtle):

```
<https://www.purl.org/SomeOntology> rdf:type owl:Ontology ;
    dcterms:creator <https://orcid.org/0000-0000-0000-0000> .
```

Example 2 (text/turtle):

```
<https://www.purl.org/SomeOntology> dcterms:creator <https://orcid.org/0000-0000-0000-0000>.

<https://orcid.org/0000-0000-0000-0000> rdf:type foaf:Person ;
    foaf:firstName "Max" ;
    foaf:lastName "Muster" .
```

Alternative properties:

- <http://purl.org/dc/elements/1.1/creator>
- <https://schema.org/creator>
- <http://purl.org/pav/createdBy>
- <http://www.w3.org/ns/prov#wasAttributedTo>
- <https://schema.org/accountablePerson>
- <https://schema.org/author>

SHACL validation rules:

- `sh:minCount 1`
- `sh:nodeKind sh:IRI`

3.1.5 Ontology version IRI

Since the PURL of an ontology usually only points to the latest version of an ontology, each ontology must state its version IRI. A version IRI is a persistent identifier for a version of an ontology and is used to reliably retrieve this earlier version of the ontology. We recommended to

use [semantic versioning](#) or OBO style date-based versioning [3].

Recommended property: <http://www.w3.org/2002/07/owl#versionIRI>

Example 1 (text/turtle):

```
<https://www.purl.org/SomeOntology> rdf:type owl:Ontology ;
    owl:versionIRI <https://www.purl.org/SomeOntology/1.0.0> .
```

Example 2 (text/turtle)

```
<https://www.purl.org/SomeOntology> rdf:type owl:Ontology ;
    owl:versionIRI <https://www.purl.org/SomeOntology/2019-12-31> .
```

Example 3 (text/turtle):

```
<http://purl.obolibrary.org/obo/pco.owl> rdf:type owl:Ontology ;
    owl:versionIRI <http://purl.obolibrary.org/obo/pco/releases/2021-05-03/pco.owl> .
```

Alternative properties: n/a

SHACL validation rules:

- `sh:maxCount 1`
- `sh:minCount 1`
- `sh:nodeKind sh:IRI`

3.1.6 Ontology creation date

The ontology must state the date of its first creation.

Recommended property: <http://purl.org/dc/terms/created>

Example (text/turtle):

```
<https://www.purl.org/SomeOntology> rdf:type owl:Ontology ;
    dcterms:created "2020-11-19T00:00:00"^^xsd:dateTime.
```

Alternative properties:

- <http://purl.org/pav/createdOn>
- <https://schema.org/dateCreated>
- <http://www.w3.org/ns/prov#generatedAtTime>

SHACL validation rules:

- `sh:xone ( [sh:datatype xsd:dateTimeStamp ;] [sh:datatype xsd:dateTime ;] [sh:datatype xsd:date ;] [sh:datatype xsd:gYearMonth ;] [sh:datatype xsd:gYear ;] ) ;`
- `sh:maxCount 1`
- `sh:minCount 1`

3.1.7 Ontology abstract

The ontology must describe its own contents and scope with a few words or sentences in order to inform human users what the ontology tries to accomplish. The abstract is displayed in the [TIB Terminology Service ontology list](#) and on the landing page of each ontology - we therefore recommend to keep it short. You can also provide abstracts in several languages (as an `rdf:langString`). You may provide only one abstract per language. Each abstract may be up to 500 characters long including spaces. We do not recommend to use markups (html,

markdown) in the abstract since these are not supported by [TIB Terminology Service](#). If you want to include a longer text about the ontology, you should include a description [3.3.1 Ontology description](#). For an extensive discussion about further aspects of the ontology (e.g. ontology creation, ontology use, ontology structure, etc.) we recommend the publication of a detailed documentation (cf. [3.1.9 Ontology documentation](#)) or traditional academic article (cf. [3.2.13 Related resources](#)).

Recommended property: <http://purl.org/dc/terms/abstract>

Example (text/turtle):

```
<https://www.purl.org/SomeOntology> rdf:type owl:Ontology ;
    dct:abstract "SomeOntology defines a range of classes and properties which can be applied to do
stuff in a particular domain."@en , "SomeOntology definiert eine Reihe von Klassen und Properties, die in
einer bestimmten Domäne verwendet werden können, um etwas damit zu tun."@de .
```

Alternative properties:

- <https://schema.org/abstract>

SHACL validation rules:

- `sh:datatype rdf:langString`
- `sh:maxLength 500`
- `sh:minCount 1`
- `sh:uniqueLang true`

3.1.8 Ontology issue tracker

The ontology must point to the issue tracker of its own development environment so that others may report bugs or suggestions to the developers. Ideally, the development process of an ontology is open and takes place on platforms like GitLab.com that allow for version management with version control software like git.

Recommended property: <http://usefulinc.com/ns/doap#bug-database>

Example (text/turtle):

```
<https://www.purl.org/SomeOntology> rdf:type owl:Ontology ;
    doap:bug-database <https://github.com/SomeOrganisation/SomeOntology/issues> .
```

Alternative properties: n/a

SHACL validation rules:

- `sh:maxCount 1`
- `sh:minCount 1`
- `sh:nodeKind sh:IRI`

3.1.9 Ontology documentation

To familiarize interested others with the concepts and scope of your ontology, some kind of documentation must be provided. This is usually an external online document that should be referenced by the ontology via an IRI or more persistently with a PID. A relatively easy way to do this is to use the tool Widoco [\[4\]](#), [\[5\]](#). It generates the documentation from the ontology and a resulting html-document can be published, for example with [GitLab Pages](#). This auto-generated document can contain customized passages that provide users with a deeper understanding of the ontology. There are, however, other forms of ontology documentations: scientific articles, well-curated README files in repositories, Wikis etc. Here, we recommend to choose the form which is most easily attainable by your working group or project. Here are some examples for ontology documentation:

- PROV Ontology: <https://www.w3.org/TR/2013/REC-prov-o-20130430/>
- GND Ontology: <https://d-nb.info/standards/elementset/gnd>
- Gene Ontology - Relations: <https://geneontology.org/docs/ontology-relations/>
- CHEBI User Manual: https://docs.google.com/document/d/1_w-DwBdCCOh1gMeeP6yqGzcnkpbHYOa3AGSODE5epcg/

- CHEBI Developer Manual: <https://docs.google.com/document/d/11G6SmTtQRQYFT7I9h5K0faUHiAaekcLeOweMOOTIpME/>

The documentation of an ontology offers a lot of room to describe the ontology and provide detailed information about it to its readers. In fact, you may wonder whether it would be sufficient to provide a well-structured and well-readable text document - containing all the metadata suggested in this guide. Here, our answer is a clear no: Providing the metadata of your ontology in a machine-readable format, makes them much more re-usable (cf. sections [0. Executive summary](#) and [1.2 Why ontology metadata](#)).

Recommended property: <http://www.loc.gov/premis/rdf/v3/documentation>

Example (text/turtle):

```
<https://www.purl.org/SomeOntology> rdf:type owl:Ontology ;  
    premis:documentation <https://www.purl.org/SomeOntology/Documentation> .
```

Alternative properties: n/a

SHACL validation rules:

- `sh:maxCount 1`
- `sh:minCount 1`
- `sh:nodeKind sh:IRI`

3.2 Recommended Metadata

In addition to mandatory metadata, we recommend providing a number of further helpful metadata.

3.2.1 Ontology contributor(s)

If you had help in developing the ontology, you should indicate this by listing contributors so that all participants receive proper credit for their efforts. If an ontology is developed by a larger group, it is recommended to give the organization or project identifier as the creator and list individual persons as contributors. Persons and organisations should be listed via PID, e.g. ORCID or ROR.

Recommended property: <http://purl.org/dc/terms/contributor>

Example 1 (text/turtle):

```
<https://www.purl.org/SomeOntology> rdf:type owl:Ontology ;
    dcterms:contributor <https://orcid.org/0000-0000-0000-0000> .
```

Example 2 (text/turtle):

```
<https://www.purl.org/SomeOntology> rdf:type owl:Ontology ;
    dcterms:contributor <https://orcid.org/0000-0000-0000-0000> .

<https://orcid.org/0000-0000-0000-0000> rdf:type foaf:Person ;
    foaf:firstName "Max" ;
    foaf:lastName "Muster" .
```

Alternative properties:

- <http://purl.org/dc/elements/1.1/contributor>
- <https://schema.org/contributor>
- <http://purl.org/pav/contributedBy>

SHACL validation rules:

- `sh:nodeKind sh:IRI`

3.2.2 Ontology funder

The development of an ontology may rely on external funding and take place in a third-party funding project. Since funding institutions usually want to be credited, we highly recommend mentioning them in your ontology. The best way to do so is by referring to their [ROR ID](#).

Recommended property: <https://schema.org/funder>

Example (text/turtle):

```
<https://www.purl.org/SomeOntology> rdf:type owl:Ontology ;
    sdo:funder <https://ror.org/018mejw64> .
```

Alternative properties:

- <http://rdf-vocabulary.ddialliance.org/discovery#fundedBy>
- <http://xmlns.com/foaf/0.1/fundedBy>

SHACL validation rules:

- `sh:nodeKind sh:IRI`

3.2.3 Ontology funding

In addition to referencing the funding institution, you may be required to point to the specific grant that enables the work on the ontology. We recommend to provide this information as an IRI, ideally a PID. We are aware, though, that PIDs for grants are not common, yet. We therefore also accept this information in the form of acknowledgement statements (`rdf:langString`) containing the grant number provided by the funding institution.

Recommended property: <https://schema.org/funding>

Example 1 (text/turtle):

```
<https://www.purl.org/SomeOntology> rdf:type owl:Ontology ;
    sdo:funding <https://doi.org/00.00000/000000000> .
```

Example 2 (text/turtle):

```
<https://www.purl.org/SomeOntology> rdf:type owl:Ontology ;
    sdo:funding "The authors (Some Ontology Workgroup) would like to thank the Government of Some Country or Some Other Funding Institution for their funding and support within Some Funding Program (project number 123456789)."@en .
```

Alternative properties: n/a

SHACL validation rules:

- `sh:nodeKind sh:IRI`

3.2.4 Ontology audience description

Defining the target group of your ontology might be useful information for others when evaluating if and how they could reuse your ontology. You should describe the intended audience of the ontology. The audience description should be a short, language-tagged text.

Recommended property: <http://usefulinc.com/ns/doap#audience>

Examples (text/turtle):

```
<https://www.purl.org/SomeOntology> rdf:type owl:Ontology ;
    doap:audience "This ontology is intended for researchers in derivational morphology, a branch of linguistics. [...]"@en .
```

See also [audience description of NMRC at TIB Terminology Service](#) or the [audience description of CROPUSAGE at AgroPortal](#).

Alternative properties:

- <https://schema.org/audience>
- <http://purl.org/dc/terms/audience>

SHACL validation rules:

- `sh:datatype rdf:langString`
- `sh:maxCount 1`

3.2.5 Ontology subject(s)

We recommend tagging the ontology with a subject from a controlled vocabulary to indicate which domain it belongs to or which subject it deals with. This information is helpful for terminology service and ontology registry providers: Subject tags from controlled vocabularies can be mapped to other controlled vocabularies that are the basis of filters and browsing functionalities of such services. Subject tags can help to make your ontology better findable and available for a wider audience.

Recommended property: <http://purl.org/dc/terms/subject>

Recommended controlled vocabularies:

- <https://purl.org/linsearch>
- <https://terminology.tib.eu/ts/ontologies/bk>
- <https://terminology.tib.eu/ts/ontologies/dfgfo>
- <https://explore.gnd.network/> (remember: the identifiers need to start with <http://d-nb.info/gnd/>!)

Example (text/turtle):

```
<https://www.purl.org/SomeOntology> rdf:type owl:Ontology ;
    dcterms:subject <https://d-nb.info/gnd/4067537-3>, <https://d-nb.info/gnd/4070177-3>,
    <http://uri.gbv.de/terminology/bk/42.15>, <https://github.com/tibonto/dfgfo/201-03>.
```

The [TIB Terminology Service](#) applies [LinSearch](#) labels for subject values, e.g.:

```
<https://www.purl.org/SomeOntology> dcterms:subject "Chemistry"^^xsd:string .
```

Alternative properties:

- http://purl.obolibrary.org/obo/IAO_0000136
- <https://schema.org/about>
- <https://schema.org/keywords>
- <http://www.w3.org/ns/dcat#keyword>
- <http://purl.org/dc/terms/coverage>

SHACL validation rules:

- `sh:nodeKind sh:IRI`

3.2.6 Ontology annotation language(s)

If your ontology does not only provide formal semantics but also multi-lingual annotations for entities, you should provide information about the ontology annotation languages (e.g. for term labels, term definitions, etc.) in its metadata. At least one language should be provided, since at least one set of annotations is expected in a well-documented ontology. You should only claim that the ontology uses an annotation language, if all ontology elements or an extensive part of the ontology is annotated in that language.

Recommended property: <http://purl.org/dc/terms/language>

Recommended controlled vocabulary: <http://id.loc.gov/vocabulary/iso639-2>

Example (text/turtle):

```
<https://www.purl.org/SomeOntology> rdf:type owl:Ontology ;
    dcterms:language <http://id.loc.gov/vocabulary/iso639-2/eng>, <http://id.loc.gov/vocabulary/iso639-2/tgl> .
```

Alternative properties:

- <https://schema.org/inLanguage>

SHACL validation rules:

- `sh:nodeKind sh:IRI`
- `sh:pattern "(^http://id.loc.gov/vocabulary/iso639-2/[a-z]{3}$|^https://id.loc.gov/vocabulary/iso639-2/[a-z]{3}$)"`

3.2.7 Applied logical framework

You should state which logical framework the ontology applies. The information can be given as a text, referring to the Semantic Web Standard (e.g. 'OWL 2') and possibly the OWL profile (e.g. 'OWL 2 EL profile').

Recommended property: <https://w3id.org/mod#hasFormalityLevel>

Example (text/turtle):

```
<https://www.purl.org/SomeOntology> rdf:type owl:Ontology ;
    mod:hasFormalityLevel "OWL version 2, EL profile"@en .
```

Alternative properties: n/a

SHACL validation rules:

- `sh:datatype rdf:langString`
- `sh:maxCount 1`

3.2.8 Ontology serialization/ file format

You should state the ontology's serialization/ file format. The value should be provided as an IRI from the Media Types list of the Internet Assigned Number Authority (IANA) [6] or from the W3C resource Unique URIs for File Formats [7].

Recommended property: <https://w3id.org/mod#hasSyntax>

Recommended controlled vocabularies:

- <https://www.w3.org/ns/formats>
- <https://www.iana.org/assignments/media-types/media-types>

In order to publish your ontology on the the [TIB Terminology Service](#) you must provide the ontology as one of the following media types:

- <https://www.iana.org/assignments/media-types/application/rdf+xml> resp. http://www.w3.org/ns/formats/RDF_XML
- <https://www.iana.org/assignments/media-types/text/turtle> resp. <http://www.w3.org/ns/formats/Turtle>
- OBO format (cf. <http://purl.obolibrary.org/obo/oboformat/spec.html>)

Currently, there is no registered media type for obo format.

Example (text/turtle):

```
<https://www.purl.org/SomeOntology> rdf:type owl:Ontology ;
    mod:hasSyntax <http://www.w3.org/ns/formats/Turtle> .
```

Alternative properties:

- <http://purl.org/dc/terms/format>
- <http://purl.org/dc/elements/1.1/format>
- <https://schema.org/encodingFormat>
- <http://omv.ontoware.org/2005/05/ontology#hasOntologySyntax>

SHACL validation rules:

- value must be from a defined list
- `sh:maxCount 1`
- `sh:nodeKind sh:IRI`

3.2.9 Ontology status

You should declare the current maintenance status of the ontology. You could use an English label to do so. We recommend using one of the values suggested by the OBO Foundry [8]. Ontologies are prone to link rot and sometimes left abandoned. If you cannot keep up the work on an ontology, this needs to be documented: Is your ontology retired? Is it still a draft? Or is it maintained by an active community?

Recommended property: <http://purl.org/ontology/bibo/status>

Recommended controlled vocabulary: <https://obofoundry.org/docs/OntologyStatus.html>

Example (text/turtle):

```
<https://www.purl.org/SomeOntology> rdf:type owl:Ontology ;
    bibo:status "inactive"@en .
```

Alternative properties:

- <https://schema.org/creativeWorkStatus>

SHACL validation rules:

- `sh:datatype rdf:langString`
- `sh:maxCount 1`

3.2.10 Ontology code repository

A code repository (e.g. on [GitLab.com](https://gitlab.com) or [GitHub.com](https://github.com)) should be the development environment of your ontology. It should contain the source code of your ontology but may also be used to host your documentation, your issue tracker and may contain related resources like different distributions of your ontology, versions of your ontology, application examples, competency questions, discussions and decisions - in short: the entire history of your ontology should be located in a code repository.

Recommended property: <http://usefulinc.com/ns/doap#repository>

Example (text/turtle):

```
<https://www.purl.org/SomeOntology> rdf:type owl:Ontology ;
    doap:repository <https://github.com/SomeOrganisation/SomeOntology> .
```

Alternative properties: n/a

SHACL validation rules:

- `sh:maxCount 1`
- `sh:nodeKind sh:IRI`

3.2.11 Ontology distributions/ products

There are different serializations available for ontologies but not all are parsable by any system: For example an ontology tool may be well prepared for rdf/xml but not so much for json-ld. It may help users of your ontology, if you provided IRIs to available alternative serializations or distributions of your ontology.

Recommended property: <http://www.w3.org/ns/dcat#distribution>

Example (text/turtle):

```
<https://www.purl.org/SomeOntology> rdf:type owl:Ontology;
    dcat:distribution <https://www.purl.org/SomeOntology.owl>, <https://www.purl.org/SomeOntology.json>,
    <https://www.purl.org/SomeOntology.ttl> .
```

Alternative properties:

- <https://schema.org/distribution>
- <http://purl.org/dc/terms/hasFormat>
- <http://purl.org/dc/terms/isFormatOf>
- <https://schema.org/associatedMedia>
- <https://schema.org/encoding>

SHACL validation rules:

- `sh:nodeKind sh:IRI`

3.2.12 Application example

How to use an ontology is often helpfully demonstrated by application examples and visualizations that give a glimpse about how the ontology can be used to structure actual data. If you have such application examples, these might be better found, if you link to them from the ontology. These application examples could be part of the ontology documentation, formal serializations applying the ontology or even technical applications that make use of the ontology.

Recommended property: <http://purl.org/vocab/vann/example>

Example (text/turtle):

```
<http://www.w3.org/ns/dcat> rdf:type owl:Ontology ;
  vann:example <https://www.w3.org/TR/vocab-dcat-2/#collection-of-examples> .
```

Alternative properties: n/a

SHACL validation rules:

- `sh:nodeKind sh:IRI`

3.2.13 Related resource(s)

Resources related to the ontology could be publications discussing the ontology, applications of the ontology or mapping files (see also section [3.3.17 Alignments/ mappings](#)). If any such resources exist, you should mention them in the ontology. You may also point to other related resources, for example resources that are cited or were otherwise used to inform the ontology. We recommend using a permanent identifier, such as a DOI, of the referenced resource.

Recommended property: <http://purl.org/dc/terms/references>

Example:

```
<https://www.purl.org/SomeOntology> rdf:type owl:Ontology ;
  dcterms:references <http://doi.org/10.0000/some0.ontolo> .
```

Alternative properties:

- <https://schema.org/citation>

SHACL validation rules:

- `sh:nodeKind sh:IRI`

3.2.14 Citation suggestion

You could give information on how you would like others to cite your ontologies. You may even provide structured metadata ready for import into citation management tools like Bibtex etc. Currently, there is no citation standard considering the citation of ontologies. We therefore recommend to provide at least the creators, the year of publication, the ontology title and to give the versionIRI as the access URL. In addition, you could also provide references to citable publications (cf. section [3.2.13 Related resources](#)).

Recommended property: <http://purl.org/dc/terms/bibliographicCitation>

Example (text/turtle):

```
<https://www.purl.org/SomeOntology> rdf:type owl:Ontology ;
  dcterms:bibliographicCitation "Last name, first name; last name, first name et al. (YYYY): Ontology
  title. URL: https://w3id.org/SomeOntology/1.1.6 (last accessed: DD.MM.YYYY)"^^@en .
```

Example (text/turtle):

```
<https://www.purl.org/SomeOntology> rdf:type owl:Ontology;
    dcterms:bibliographicCitation "@misc{SomeOntology, title = \"Some Ontology\", author = \"Last name,
first name and Last name, first name\", url = \"https://w3id.org/SomeOntology/1.1.6\", year =
\"YYYY\",}""^xsd:string .
```

Alternative properties: n/a

SHACL validation rules: n/a

3.2.15 Ontology sources (derived from)

You should specify whether and from which other ontologies your ontology has been derived. The reference should be made by an owl:versionIRI of the ontology from which the current ontology has been derived.

Recommended property: <http://purl.org/pav/derivedFrom>

Example (text/turtle):

```
<https://www.purl.org/SomeOntology> rdf:type owl:Ontology;
    pav:derivedFrom <https://www.purl.org/SomeOtherOntology> .
```

Alternative properties:

- <http://www.w3.org/ns/prov#wasDerivedFrom>
- <https://schema.org/isBasedOn>

SHACL validation rules:

- `sh:nodeKind sh:IRI`

3.2.16 Ontology root classes

You should explicitly declare the ontology's preferred root classes. Display tools like the [TIB Terminology Service](#) and other OLS-based services [9], [10] can pick specific, user-defined classes for rendering the ontology class hierarchy. This is especially helpful, when an ontology imports a lot of classes from other ontologies. The respective classes need to be provided via their identifier. This information should best be provided or defined by the ontology maintainers or engineers.

Recommended property: http://purl.obolibrary.org/obo/IAO_0000700

Example (text/turtle):

```
<https://www.purl.org/SomeOntology> rdf:type owl:Ontology ;
    obo:IAO_0000700 <https://www.purl.org/SomeOntology/class12345654544545> .
```

Alternative properties:

- <http://rdfs.org/ns/void#rootResource>

SHACL validation rules:

- `sh:nodeKind sh:IRI`

3.3 Optional metadata

The metadata of an ontology can contain much more information than what has been suggested in sections [3.1 Mandatory metadata](#) and [3.2 Recommended metadata](#) so far. Perhaps, you will also find the following categories useful or informative enough to include them into your metadata.

At [TIB Terminology Service](#) all additional metadata will be displayed on the ontology landing page if available.

3.3.1 Ontology description

If you want to provide more information about the ontology than the few characters that fit into the [ontology abstract](#), you may also provide a longer descriptive text. In a longer description you may, for example, provide more information about how and why the ontology was created, where it is used, how it will be updated and the like. For extensive discussions about further aspects of the ontology (e.g. ontology creation, ontology use, ontology structure, etc.) we recommend the publication of a detailed documentation (cf. [3.1.9 Ontology documentation](#)) or traditional academic article (cf. [3.2.13 Related resources](#)), instead of an ontology description.

Recommended property: <https://schema.org/description>

Example (text/turtle):

```
<https://www.purl.org/SomeOntology> rdf:type owl:Ontology;  
    sdo:description "Free text describing the ontology and how it came to be. Basically everything you  
    might want to add and which does not fit into the abstract."@en .
```

Alternative properties:

- <http://purl.org/dc/terms/description>
- <http://purl.org/dc/elements/1.1/description>

SHACL validation rules:

- `sh:datatype rdf:langString`
- `sh:nodeKind sh:Literal`
- `sh:uniqueLang true`

3.3.2 Alternative ontology title

An alternative title for the ontology may provided, for example a former working title the ontology has been known by.

Recommended property: <http://purl.org/dc/terms/alternative>

Example (text/turtle):

```
<https://www.purl.org/SomeOntology> rdf:type owl:Ontology ;  
    dcterms:alternative "My Ontology"@en .
```

Alternative properties:

- <https://schema.org/alternateName>
- <https://schema.org/alternativeHeadline>

SHACL validation rules:

- `sh:datatype rdf:langString`

3.3.3 Alternative ontology prefix/ acronym

An alternative prefix for the ontology.

Recommended property: <http://identifiers.org/idot/alternatePrefix>

Example (text/turtle):

```
<https://www.purl.org/SomeOntology> rdf:type owl:Ontology ;  
  idot:alternatePrefix "mo"^^xsd:string .
```

Alternative properties: n/a

SHACL validation rules:

- `sh:datatype xsd:string`

3.3.4 Related version/ version history

When ontologies are developed on collaborative development platforms with version management software like [GitLab.com](https://gitlab.com) or [GitHub.com](https://github.com), it is relatively easy to maintain and keep all development and release versions of an ontology. Ontologies may therefore have multiple resolvable version IRIs. The different versions can relate to each other in different ways as discussed in the following sections.

3.3.4.1 Prior ontology version

An ontology may point back to one or more versions that were valid before the current version.

Recommended property: <http://www.w3.org/2002/07/owl#priorVersion>

Example (text/turtle):

```
<https://www.purl.org/SomeOntology> rdf:type owl:Ontology ;  
  owl:priorVersion <https://www.purl.org/SomeOntology/1.1.6> .
```

Alternative properties:

- <http://www.w3.org/ns/adms#prev>
- <http://purl.org/pav/previousVersion>
- <http://www.w3.org/ns/prov#wasRevisionOf>
- <http://purl.org/dc/terms/replaces>

SHACL validation rules:

- `sh:nodeKind sh:IRI`

3.3.4.2 Related ontology version

A related resource of which the described resource is a version, edition, or adaptation. The contents of related versions of an ontology are not identical.

Recommended property: <http://purl.org/dc/terms/hasVersion>

Example (text/turtle):

```
<https://www.purl.org/SomeOntology> rdf:type owl:Ontology ;  
  dcterms:hasVersion <https://www.purl.org/SomeOntology/1.1.6> .
```

Alternative properties:

- <http://purl.org/pav/hasCurrentVersion>
- <https://schema.org/version>

SHACL validation rules:

- `sh:nodeKind sh:IRI`

3.3.4.3 Compatible ontology version

For application developers and other users, it may be useful to know whether a new version of your ontology is backward compatible with an older version of your ontology.

Recommended property: <http://www.w3.org/2002/07/owl#backwardCompatibleWith>

Example (text/turtle):

```
<https://www.purl.org/SomeOntology> rdf:type owl:Ontology ;  
  owl:versionIRI <https://www.purl.org/SomeOntology/1.1.7> ;  
  owl:backwardCompatibleWith <https://www.purl.org/SomeOntology/1.1.6> .
```

Alternative properties: n/a

SHACL validation rules:

- `sh:nodeKind sh:IRI`

3.3.4.4 Incompatible ontology version

For application developers and other users, it may be useful to know whether a new version of your ontology is incompatible with an older version of your ontology.

Recommended property: <http://www.w3.org/2002/07/owl#incompatibleWith>

Example (text/turtle):

```
<https://www.purl.org/SomeOntology> rdf:type owl:Ontology ;  
  owl:versionIRI <https://www.purl.org/SomeOntology/1.1.7> ;  
  owl:incompatibleWith <https://www.purl.org/SomeOntology/1.1.6> .
```

Alternative properties: n/a

SHACL validation rules:

- `sh:nodeKind sh:IRI`

3.3.5 Social media

If you toot about or otherwise promote your ontology on social media, it might be interesting for users to see on which platforms you are having accounts to follow your updates.

Recommended property: <http://xmlns.com/foaf/0.1/holdsAccount>

Example (text/turtle):

```
<https://www.purl.org/SomeOntology/1.1.7> rdf:type owl:Ontology ;  
  foaf:holdsAccount "myOnto@ontology.social"^^xsd:string .
```

Alternative properties: n/a

SHACL validation rules: n/a

3.3.6 KOS type

You may want to additionally classify your ontology with a controlled value from the NKOS type vocabulary [11].

Recommended property: <http://purl.org/dc/terms/type>

Recommended controlled vocabulary: <https://nkos.dublincore.org/nkostype/nkostype.rdf>

Example (text/turtle):

```
<https://www.purl.org/SomeOntology/1.1.7> rdf:type owl:Ontology ;
    dcterms:type <http://w3id.org/nkos/nkostype#ontology> .
```

Alternative properties: n/a

SHACL validation rules:

- value must be from a defined list
- `sh:maxCount 1`
- `sh:nodeKind sh:IRI`

3.3.7 Example ontology identifier

Give an example for identifiers used in your ontology.

Recommended property: <http://identifiers.org/ident/exampleIdentifier>

Example (text/turtle):

```
<https://www.purl.org/SomeOntology> rdf:type owl:Ontology ;
    idot:exampleIdentifier <https://www.purl.org/SomeOntology/1234-XY> .
```

Alternative properties: n/a

SHACL validation rules:

- `sh:nodeKind sh:IRI`

3.3.8 Ontology identifier pattern

Identifiers for ontology classes, properties and individuals usually follow one specific pattern that can be summarized with a regular expression. In registries like [Bioregistry](https://bioregistry.org/) this is usually provided as part of the ontology metadata.

Recommended property: <http://identifiers.org/ident/identifierPattern>

Example (text/turtle):

```
<https://www.purl.org/SomeOntology> rdf:type owl:Ontology ;
    idot:identifierPattern "^\\d{4}-[A-Z]{2}$"^^xsd:string .
```

Alternative properties:

- <https://bioregistry.io/schema/#0000008>

SHACL validation rules:

- `sh:datatype xsd:string`
- `sh:maxCount 1`

3.3.9 Ontology homepage

The official homepage of your ontology. This is not necessarily the documentation website, but could be a general info page, which may also include documentation about the ontology.

Recommended property: <http://xmlns.com/foaf/0.1/homepage>

Example (text/turtle):

```
<https://www.purl.org/SomeOntology> rdf:type owl:Ontology;
  foaf:homepage <https://www.example.com/SomeOntologyInfo> .
```

Alternative properties:

- <http://xmlns.com/foaf/0.1/page>

SHACL validation rules:

- `sh:maxCount 1`
- `sh:nodeKind sh:IRI`

3.3.10 Ontology publisher

The official publisher of the ontology. This might be the institution you are affiliated to. We recommend to refer to the institution via PIDs (e.g. [ROR](#), [ISNI](#), [GND](#) ID).

Recommended property: <http://purl.org/dc/terms/publisher>

Example (text/turtle):

```
<https://www.purl.org/SomeOntology> rdf:type owl:Ontology ;
  dcterms:publisher <https://isni.org/isni/0000000121746694> .
```

Alternative properties:

- <http://purl.org/dc/elements/1.1/publisher>
- <https://schema.org/publisher>

SHACL validation rules:

- `sh:nodeKind sh:IRI`
- `sh:pattern "https://d-nb.info/gnd/(|{1[012]?[0-9]{7}[0-9X]}|[47][0-9]{6}-[0-9]|[1-9][0-9]{0,7}-[0-9X]}|3[0-9]{7}[0-9X]))$"`
- `sh:pattern "https://isni.org/isni/[0]{4}[0-9]{4}[0-9]{4}[0-9]{3}[0-9X]"`
- `sh:pattern "https://ror.org/([a-z0-9]{9})"`

3.3.11 Ontology comments

Any comments you would like to make about your ontology.

Recommended property: <http://www.w3.org/2000/01/rdf-schema#comment>

Example (text/turtle):

```
<https://www.purl.org/SomeOntology> rdf:type owl:Ontology ;
  rdfs:comment "SomeOntology does not actually exist."@en .
```

Alternative properties:

- <https://schema.org/comment>

SHACL validation rules:

- `sh:datatype rdf:langString`

3.3.12 Example ontology class

You could provide an example class that is representative for the entities described in the ontology, for example demonstrating typical term annotations, editorial notes or axiomatic statements, deprecation notes etc.

Recommended property: <http://rdfs.org/ns/void#exampleResource>

Example (text/turtle):

```
<https://www.purl.org/SomeOntology> rdf:type owl:Ontology ;  
    void:exampleResource <https://www.purl.org/SomeClass> .
```

Alternative properties:

- <http://www.w3.org/2004/02/skos/core#example>

SHACL validation rules:

- `sh:nodeKind sh:IRI`

3.3.13 Ontology mailing list

Communication about the ontology may take place over a mailing list, e.g. regarding updates. If this is your way to communicate, the info should be part of your ontology.

Recommended property: <http://usefulinc.com/ns/doap#mailing-list>

Example (text/turtle):

```
<https://www.purl.org/SomeOntology> rdf:type owl:Ontology ;  
    doap:mailing-list <someontology@list.de> .
```

Alternative properties: n/a

SHACL validation rules:

- `sh:maxCount 1`

3.3.14 Ontology logo/ depiction/ related visualizations

Links to official ontology logo or other visualizations/ diagrams of ontology elements online, e.g. WebVOWL visualizations and other graph views.

Recommended property: <http://xmlns.com/foaf/0.1/logo>

Example (text/turtle):

```
<https://www.purl.org/SomeOntology> rdf:type owl:Ontology ;  
    foaf:logo <https://www.example.com/MyOntoLogo.jpg>
```

Alternative properties:

- <https://schema.org/logo>
- <http://xmlns.com/foaf/0.1/depiction>
- <https://w3id.org/mod#depiction>
- <https://schema.org/image>

SHACL validation rules:

- `sh:nodeKind sh:IRI`

3.3.15 Alternative identifiers of the ontology

If your ontology has been published at an archive, you may want to declare these alternative identifiers in the ontology metadata. You should list here alternative URIs that may be used to identify your ontology. The identifier used as the base URI of the ontology should be

provided with <http://purl.org/vocab/vann/preferredNamespaceUri> instead (cf. 3.3.20 Preferred ontology namespace).

Recommended property: <http://purl.org/dc/terms/identifier>

Example (text/turtle):

```
<https://www.purl.org/SomeOntology> rdf:type owl:Ontology ;  
    dcterms:identifier <https://doi.org/10.5281/zenodo.0000000> .
```

Alternative properties:

- <http://purl.org/ontology/bibo/doi>
- <https://schema.org/identifier>
- <http://purl.org/dc/elements/1.1/identifier>

SHACL validation rules:

- `sh:nodeKind sh:IRI`

3.3.16 Development environment

The software that was used to create the ontology.

Recommended property: <http://purl.org/pav/createdWith>

Example (text/turtle):

```
<https://www.purl.org/SomeOntology> rdf:type owl:Ontology ;  
    pav:createdWith "Protégé 4.8.8"^^xsd:string .
```

Alternative properties:

- <https://w3id.org/mod#createdWith>

SHACL validation rules: n/a

3.3.17 Alignments/ mappings

3.3.17.1 Aligned resources

The ontology may indicate to which other resource(s) it has been aligned or contains equivalences to.

Recommended property: <https://w3id.org/mod#hasEquivalencesWith>

Example (text/turtle):

```
<https://www.purl.org/SomeOntology> rdf:type owl:Ontology ;  
    mod:hasEquivalencesWith <https://www.purl.org/SomeOtherOntology> .
```

Alternative properties:

- <http://w3id.org/nkos#alignedWith>

SHACL validation rules:

- `sh:nodeKind sh:IRI`

3.3.17.2 Alignment files/ mapping files

Alignments and mappings may not be included in the ontology document itself, but in a related resource (e.g. when following the SSSOM paradigm [12]). Accordingly, we recommend to treat mapping sets like related resources. The mapping set should be referenced via a PID or

a version IRI.

Recommended property: <http://purl.org/dc/terms/references>

Example (text/turtle):

```
<https://www.purl.org/SomeOntology> rdf:type owl:Ontology ;
    dcterms:references <https://www.purl.org/SomeOntology/MappingSet1>.
```

Alternative properties: n/a

SHACL validation rules:

- `sh:nodeKind sh:IRI`

3.3.18 Competency questions

Which competency questions does the ontology address? For which use case was it developed?

Recommended property: <https://w3id.org/mod#competencyQuestion>

Example 1 (text/turtle)

```
<https://www.purl.org/SomeOntology> rdf:type owl:Ontology ;
    mod:competencyQuestion "Who developed an ontology?"@en .
```

Example 2 (text/turtle):

```
<https://www.purl.org/SomeOntology> rdf:type owl:Ontology ;
    mod:competencyQuestion <https://www.example.com/SomeOntologyInfo/Questions#Q1>
```

Alternative properties: n/a

SHACL validation rules:

- `sh:xone ([sh:nodeKind sh:IRI ;] [sh:datatype rdf:langString ;])`

3.3.19 Applied methodology

A name or description of the steps taken to develop the ontology. This should describe the overall organisation of the ontology development process.

Recommended property: <http://omv.ontoware.org/2005/05/ontology#usedOntologyEngineeringMethodology>

Example (text/turtle):

```
<https://www.purl.org/SomeOntology> rdf:type owl:Ontology ;
    omv:usedOntologyEngineeringMethodology "Methontology" .
```

Alternative properties: n/a

SHACL validation rules: n/a

3.3.20 Preferred ontology namespace

The preferred namespace URI of an ontology is the URI which is to be used when referencing its terms.

Recommended property: <http://purl.org/vocab/vann/preferredNamespaceUri>

Example (text/turtle):

```
<https://www.purl.org/SomeOntology> rdf:type owl:Ontology ;
  vann:preferredNamespaceUri <https://w3id.org/Example/SomeOntology> .
```

Alternative properties: n/a

SHACL validation rules:

- `sh:maxCount 1`
- `sh:nodeKind sh:IRI`

3.3.21 Ontology issue date

The date the ontology was officially published.

Recommended property: <http://purl.org/dc/terms/issued>

Example (text/turtle):

```
<https://www.purl.org/SomeOntology> rdf:type owl:Ontology ;
  dcterms:issued "2023-11-21"^^xsd:dateTime .
```

Alternative properties:

- <https://schema.org/dateIssued>
- <https://schema.org/datePublished>

SHACL validation rules:

- `sh:xone ( [sh:datatype xsd:dateTimeStamp ;] [sh:datatype xsd:dateTime ;] [sh:datatype xsd:date ;] ) ;`
- `sh:maxCount 1`

3.3.22 Modification date

Since ontologies are updated over time, it would be helpful to provide the date when the ontology has been last modified.

Recommended property: <http://purl.org/dc/terms/modified>

Example (text/turtle):

```
<https://www.purl.org/SomeOntology> rdf:type owl:Ontology ;
  dcterms:modified "2020-11-19T00:00:00"^^xsd:dateTime .
```

Alternative properties:

- <https://schema.org/dateModified>
- <http://purl.org/pav/curatedOn>
- <http://purl.org/pav/lastUpdateOn>

SHACL validation rules:

- `sh:xone ( [sh:datatype xsd:dateTimeStamp ;] [sh:datatype xsd:dateTime ;] [sh:datatype xsd:date ;] ) ;`
- `sh:maxCount 1`

3.3.23 Textual version information

Some ontologies make use of semantic versioning and employ strings like 1.0.0 as a tag to distinguish one version of their ontology from a successor. Others employ the modification date, e.g. 2022-12-21. If you use such textual version information but do not use/have a Version URI, then we strongly recommend to also mint a version URI in which you use the textual version information as variable.

If you need to add a larger comment in natural language, you should provide the info as an `rdf:langString`, i.e. with a language tag. We do not encourage this, since such statements can most likely be expressed in more granular fashion with formal statements. Alternatively, you could add the info to the version notes (cf. [3.3.24 Version notes](#)).

Recommended property: <http://www.w3.org/2002/07/owl#versionInfo>

Example 1 (text/turtle):

```
<https://www.purl.org/SomeOntology> rdf:type owl:Ontology ;
    owl:versionInfo "2023-01-01"^^xsd:string .
```

Example 2 (text/turtle):

```
<https://www.purl.org/SomeOntology> rdf:type owl:Ontology ;
    owl:versionInfo "1.0.0"^^xsd:string .
```

Example 3 (text/turtle) :

```
<https://www.purl.org/SomeOntology> rdf:type owl:Ontology ;
    owl:versionInfo "Ontology version 1.0.0 of the subject classification in tabular format from Nov
2024."@en.
```

Alternative properties:

- <http://purl.org/pav/version>

SHACL validation rules:

- `sh:xone ([sh:datatype xsd:string ;] [sh:datatype rdf:langString ;])`
- `sh:maxCount 1`

3.3.24 Version notes

Version information may also be accompanied by a description about changes between one version of the ontology and its predecessor.

There are several ways for an automatic generation of change notes. When using GitHub and properly atomic PRs for the implementation of each change, you could copy the notes that are created when preparing a release based on tags, which will list all the PRs that have been merged since the last tag. Another approach could be to use the output of the ROBOT diff command.

Recommended property: <http://www.w3.org/ns/adms#versionNotes>

Example (text/turtle):

```
<https://www.purl.org/SomeOntology> rdf:type owl:Ontology ;
    adms:versionNotes "Changes from 2000-01-12: add annotation properties: familyName [...]"@en .
```

Alternative properties:

- <http://purl.org/vocab/vann/changes>

SHACL validation rules:

- `sh:datatype rdf:langString`
- `sh:nodeKind sh:Literal`

3.4 Tabular overview

section	Recommended property	Mandatory	Recommended	Optional	Cardinality
3.1.1 Ontology title	http://purl.org/dc/terms/title	x			1..*
3.1.2 Preferred ontology prefix/ acronym	http://purl.org/vocab/vann/preferredNamespacePrefix	x			1
3.1.3 Ontology license	http://purl.org/dc/terms/license	x			1
3.1.4 Ontology creator(s)	http://purl.org/dc/terms/creator	x			1..*
3.1.5 Ontology version IRI	http://www.w3.org/2002/07/owl#versionIRI	x			1
3.1.6 Ontology creation date	http://purl.org/dc/terms/created	x			1
3.1.7 Ontology abstract	http://purl.org/dc/terms/abstract	x			1..*
3.1.8 Ontology issue tracker	http://usefulinc.com/ns/doap#bug-database	x			1
3.1.9 Ontology documentation	http://www.loc.gov/premis/rdf/v3/documentation	x			1
3.2.1 Ontology contributor(s)	http://purl.org/dc/terms/contributor		x		0..*
3.2.2 Ontology Funder	https://schema.org/funder		x		0..*
3.2.3 Ontology funding	https://schema.org/funding		x		0..*
3.2.4 Ontology audience description	http://usefulinc.com/ns/doap#audience		x		0..1
3.2.5 Ontology subjects	http://purl.org/dc/terms/subject		x		0..*
3.2.6 Ontology annotation languages	http://purl.org/dc/terms/language		x		0..*
3.2.7 Applied logical framework	https://w3id.org/mod#hasFormalityLevel		x		0..1
3.2.8 Ontology serialization/ file format	https://w3id.org/mod#hasSyntax		x		0..1
3.2.9 Ontology status	http://purl.org/ontology/bibo/status		x		0..1
3.2.10 Ontology Code Repository	http://usefulinc.com/ns/doap#repository		x		0..1

section	Recommended property	Mandatory	Recommended	Optional	Cardinality
3.2.11 Ontology distribution/ products	http://www.w3.org/ns/dcat#distribution		x		0..*
3.2.12 Application example	http://purl.org/vocab/vann/example		x		0..*
3.2.13 Related resources	http://purl.org/dc/terms/references		x		0..*
3.2.14 Citation suggestion	http://purl.org/dc/terms/bibliographicCitation		x		0..*
3.2.15 Ontology sources/ derived from	http://purl.org/pav/derivedFrom		x		0..*
3.2.16 Ontology root classes	http://purl.obolibrary.org/obo/IAO_0000700		x		0..*
3.3.1 Ontology deswcription	https://schema.org/description			x	0..*
3.3.2 Alternative ontology title	http://purl.org/dc/terms/alternative			x	0..*
3.3.3 Alternative ontology/ prefix acronym	http://identifiers.org/idot/alternatePrefix			x	0..*
3.3.4.1 Prior ontology version	http://www.w3.org/2002/07/owl#priorVersion			x	0..*
3.3.4.2 Related ontology version	http://purl.org/dc/terms/hasVersion			x	0..*
3.3.4.3 Compatible ontology version	http://www.w3.org/2002/07/owl#backwardCompatibleWith			x	0..*
3.3.4.4 Incompatible ontology version	http://www.w3.org/2002/07/owl#incompatibleWith			x	0..*
3.3.5 Social media	http://xmlns.com/foaf/0.1/holdsAccount			x	0..*
3.3.6 KOS type	http://purl.org/dc/terms/type			x	0..1
3.3.7 Example ontology identifier	http://identifiers.org/idot/exampleIdentifier			x	0..*

section	Recommended property	Mandatory	Recommended	Optional	Cardinality
3.3.8 Ontology identifier pattern	http://identifiers.org/idot/identifierPattern			x	0..1
3.3.9 Ontology homepage	http://xmlns.com/foaf/0.1/homepage			x	0..1
3.3.10 Ontology publisher	http://purl.org/dc/terms/publisher			x	0..*
3.3.11 Ontology comments	http://www.w3.org/2000/01/rdf-schema#comment			x	0..*
3.3.12 Example ontology class	http://rdfs.org/ns/void#exampleResource			x	0..*
3.3.13 Ontology Mailing List	http://usefulinc.com/ns/doap#mailing-list			x	0..1
3.3.14 Ontology logo/depiction/related visualizations	http://xmlns.com/foaf/0.1/logo			x	0..*
3.3.15 Alternative identifiers of the ontology	http://purl.org/dc/terms/identifier			x	0..*
3.3.16 Development environment	http://purl.org/pav/createdWith			x	0..*
3.3.17.1 Aligned resources	https://w3id.org/mod#hasEquivalencesWith			x	0..*
3.3.17.2 Alignment files/ mapping files	http://purl.org/dc/terms/references			x	0..*
3.3.18 Competency questions	https://w3id.org/mod#competencyQuestion			x	0..*
3.3.19 Applied methodology	< http://omv.ontoware.org/2005/05/ontology#usedOntologyEngineeringMethodology >			x	0..*
3.3.20 Preferred ontology namespace	http://purl.org/vocab/vann/preferredNamespaceUri			x	0..1
3.3.21 Ontology issue date	http://purl.org/dc/terms/issued			x	0..1

section	Recommended property	Mandatory	Recommended	Optional	Cardinality
3.3.22 Modification date	http://purl.org/dc/terms/modified			x	0..1
3.3.23 Textual version information	http://www.w3.org/2002/07/owl#versionInfo			x	0..1
3.3.24 Textual version information	http://www.w3.org/ns/adms#versionNotes			x	0..*

3.5 Relations to related work

The following table shows the relation of the recommendations in this guide to related works. If these works recommend the same metadatum for an ontology, the corresponding cell will be marked as *true*, if the metadatum could not be identified in the source, the corresponding cell will be marked as *false*.

section	[13]	[14]	[15]	[16]	[17]	[18]	[19]	[20]	[21]	[22]	[23]
3.1.1 Ontology title	true	true	true	true	true	true	true	false	false	true	true
3.1.2 Preferred ontology prefix/ acronym	true	true	true	false	true	true	true	false	false	true	false
3.1.3 Ontology license	true	true	true	true	true	true	true	true	true	true	true
3.1.4 Ontology creator(s)	true	true	true	true	true	true	true	true	true	false	true
3.1.5 Ontology version IRI	false	false	false	false	true	true	true	true	true	false	true
3.1.6 Ontology creation date	true	true	false	false	true	true	true	true	true	false	true
3.1.7 Ontology abstract	false	true	false	true	true	true	true	false	false	true	true
3.1.8 Ontology issue tracker	true	true	false	true	false	false	true	false	false	false	true
3.1.9 Ontology documentation	false	false	false	false	false	false	false	false	true	false	false
3.2.1 Ontology contributor(s)	false	true	true	true	true	true	true	true	false	false	true
3.2.2 Ontology Funder	true	true	false	false	true	false	true	false	false	false	true
3.2.3 Ontology funding	false	false	false	false	true	false	false	false	false	false	true
3.2.4 Ontology audience description	true	true	false	true	false	false	true	false	false	false	true
3.2.5 Ontology subjects	true	true	false	true	false	false	true	false	true	true	true
3.2.6 Ontology annotation languages	true	true	false	false	false	false	true	false	false	false	true
3.2.7 Applied logical framework	false	true	false	true	false	false	true	false	false	false	false
3.2.8 Ontology serialization/ file format	true	true	false	false	false	false	true	false	false	false	true
3.2.9 Ontology status	true	false	false	true	true	true	true	true	true	true	true
3.2.10 Ontology Code Repository	false	true	false	true	false	false	true	false	false	true	false
3.2.11 Ontology distribution/ products	true	true	false	false	false	false	true	false	true	false	true
3.2.12 Application example	false	true	false	true	false	false	true	false	false	false	false
3.2.13 Related resources	true	true	false	false	true	true	true	false	true	true	true
3.2.14 Citation suggestion	false	true	false	false	true	true	true	true	false	false	true
3.2.15 Ontology sources/ derived from	true	true	false	true	true	true	true	true	true	false	true
3.2.16 Ontology root classes	false	true	false	false	false	false	true	false	false	false	false
3.3.1 Ontology description	true	true	true	true	true	true	true	false	true	true	true
3.3.2 Alternative ontology title	false	true	false	false	false	false	true	false	false	false	true
3.3.3 Alternative ontology/ prefix acronym	true	true	false	false	false	false	false	false	false	true	false
3.3.4.1 Prior ontology version	false	true	false	false	true	true	true	true	true	false	false
3.3.4.2 Related ontology version	false	true	false	false	false	false	true	false	false	false	false
3.3.4.3 Compatible ontology version	false	false	false	false	true	true	true	true	false	false	false
3.3.4.4 Incompatible ontology version	false	false	false	false	true	true	true	true	false	false	false
3.3.5 Social media	true	false	false	false	false	false	false	false	false	true	false
3.3.6 KOS type	false	false	false	false	false	false	false	false	false	false	true

section	[13]	[14]	[15]	[16]	[17]	[18]	[19]	[20]	[21]	[22]	[23]
3.3.7 Example ontology identifier	true	true	false	false	false	false	true	false	false	true	false
3.3.8 Ontology identifier pattern	true	true	false	true	false	false	true	false	false	true	false
3.3.9 Ontology homepage	true	true	false	false	false	false	true	false	false	true	false
3.3.10 Ontology publisher	true	true	true	false	true	true	true	false	false	false	true
3.3.11 Ontology comments	true	true	true	false	false	false	true	false	false	true	false
3.3.12 Example ontology class	true	true	false	false	true	false	true	false	false	false	true
3.3.13 Ontology Mailing List	true	true	false	true	false	false	true	false	false	false	false
3.3.14 Ontology logo/ depiction/ related visualizations	true	true	false	true	true	true	true	false	false	true	false
3.3.15 Alternaitve identifiers of the ontology	true	true	false	false	true	true	true	false	false	true	true
3.3.16 Development environment	true	true	false	true	false	true	true	false	false	false	false
3.3.17.1 Aligned resources	false	true	false	false	false	false	true	false	true	false	false
3.3.17.2 Alignment files/ mapping files	false	false	false	false	false	false	false	false	true	false	false
3.3.18 Competency questions	false	true	false	true	false	false	true	false	false	false	false
3.3.19 Applied methodology	true	true	false	true	false	false	true	false	false	false	false
3.3.20 Preferred ontology namespace	true	true	true	false	true	true	true	false	false	false	true
3.3.21 Ontology issue date	false	true	true	false	true	true	false	false	false	false	false
3.3.22 Modification date	false	true	true	false	true	true	true	true	false	false	true
3.3.23 Textual version information	false	true	true	false	true	true	true	true	false	true	true
3.3.24 Textual version information	false	true	true	false	false	true	false	true	false	false	false

4 Term-level Metadata

Not only the ontology itself should have rich metadata: In order to better understand the scope and purpose of an ontology term, it should be annotated as well. This, in turn, helps to assess whether the element is fit for re-use in a different context.

With *term* we refer to classes, properties and individuals. A term's annotations could include all information that tells the ontology audience what the term is about or how it may be used. In the following sections we discuss some annotations we deem mandatory, some that we would recommend and a limited number of optional ones.

4.1 Mandatory metadata

4.1.1 (Preferred) label

Each ontoloy entity should have a label in at least one natural language. A label can be either a single word, compound or other kinds of multi-word expressions. The de facto *lingua franca* in ontology development is English but there are also multi-lingual ontologies. Regardless of how many languages an ontolgy supports, the labels should always be explicitly language-tagged - even if the ontology only provides data in only one language! The label should be the preferred label of an entity.

There should only be one preferred label per entity and language.

Recommended property: <http://www.w3.org/2000/01/rdf-schema#label>

Example (text/turtle):

```
<https://www.purl.org/SomeOntologyClass> rdf:type owl:Class ;
  rdfs:label "smart grid"@en .
```

Alternative properties:

- <http://www.w3.org/2004/02/skos/core#prefLabel>
- <http://purl.org/dc/terms/title>
- <http://purl.org/dc/elements/1.1/title>
- <http://schema.org/name>
- <https://d-nb.info/standards/elementset/gnd#preferredNameForTheSubjectHeading>
- <https://physh.org/rdf/2018/01/01/core#prefLabel>

SHACL validation rules:

- `sh:minCount 1`
- `sh:datatype rdf:langString`
- `sh:uniqueLang true`

4.1.2 Definition

Each ontology term should have a short description defining it. A typical definition usually consists of two parts: a reference to a super-ordinate term and a statement naming its defining characteristics. The characteristics usually serve to distinguish the term from related terms. The definition should explicitly be language-tagged even if the ontology only serves data in one language! There may only be one definition per term and language.

Recommended property: <http://www.w3.org/2004/02/skos/core#definition>

Example (text/turtle):

```
<https://www.purl.org/SomeOntologyClass> rdf:type owl:Class ;
    skos:definition "electrical grid with information and communication technology as well as automation
mechanisms that includes a great number of decentralized electrical energy sources"@en .
```

Alternative properties:

- http://purl.obolibrary.org/obo/IAO_0000115
- <http://purl.org/dc/terms/description>
- <https://d-nb.info/standards/elementset/gnd#definition>
- <http://www.w3.org/ns/prov#definition>
- <http://purl.org/dc/elements/1.1/description>
- <http://www.geneontology.org/formats/oboInOwl#hasDefinition>
- http://emmo.info/emmo#EMMO_70fe84ff_99b6_4206_a9fc_9a8931836d84

Unaccepted properties:

- <http://www.w3.org/2000/01/rdf-schema#comment>

SHACL validation rules:

- `sh:minCount 1`
- `sh:datatype rdf:langString`
- `sh:uniqueLang true`

4.1.3 Term deprecation

Terms that shall no longer be used, must be marked with respective deprecation information. The deprecation annotations are only considered mandatory if the term is deprecated. The following sections describe which elements are needed in order to deprecate a term.

4.1.3.1 Deprecation marker

The deprecation marker is a boolean value intended to indicate the fact that a term is deprecated and should not be used any longer.

Recommended property: <http://www.w3.org/2002/07/owl#deprecated>

Example (text/turtle):

```
<https://www.purl.org/SomeOntologyClass> rdf:type owl:Class ;
    owl:deprecated true .
```

Alternative properties: n/a

SHACL validation rules:

- `sh:datatype xsd:boolean`

4.1.3.2 Obsolescence reason

For users of the ontology it may be helpful to understand why a term has been deprecated. The reason should be explained with a few words. The IAO property we recommend to use for this cause expects items from a controlled list, i.e. instances of the class [obsolescence reason specification](#) (IAO:0000225). If any of these suit your ontology, we recommend to use these items. However, we would also like to leave the possibility open to provide some free text explaining the reason for obsoleting a term.

Recommended property: [has obsolescence reason](#) (IAO:0000231)

Example (text/turtle):

```
<https://www.purl.org/SomeOntologyClass> rdf:type owl:Class ;
    obo:IAO_0000231 <http://purl.obolibrary.org/obo/IAO_0000103>.
```

Alternative properties: n/a

SHACL validation rules:

- `sh:xone ( [sh:datatype rdf:langString ;] [sh:nodeKind sh:IRI ;])`
- `sh:in ( <http://purl.obolibrary.org/obo/IAO_0000103> <http://purl.obolibrary.org/obo/IAO_0000226> <http://purl.obolibrary.org/obo/IAO_0000228> <http://purl.obolibrary.org/obo/IAO_0000229> <http://purl.obolibrary.org/obo/IAO_0000227> <http://purl.obolibrary.org/obo/OMO_0001000> )`

4.1.3.3 Term replacement

In some cases there may be a replacement term that can be used instead of an obsoleted term. The deprecated term should point to this replacement.

Recommended property: [term replaced by](#) (IAO:0100001)

Example (text/turtle):

```
<https://www.purl.org/SomeOntologyClass> rdf:type owl:Class ;
    obo:IAO_0100001 <https://www.purl.org/SomeOtherOntologyClass>.
```

Alternative properties: n/a

SHACL validation rules:

- `sh:nodeKind sh:IRI`

4.2 Recommended metadata

4.2.1 Synonyms/ alternative labels

An ontology term may have more than just one label which it can be referred by. Since synonyms are an obstacle not only to understanding but also to the search for information, an ontology is a good place to collect frequent synonyms of a term. These can be leveraged in search applications over text in order to intercept with linguistic variation that is inevitable when different communicators interact. We recommend to list only very close synonyms with the properties suggested here. Short forms, acronyms, near synonyms, related synonyms, broad synonyms, narrow synonyms etc. should be provided using a dedicated property.

Recommended property: <http://www.w3.org/2004/02/skos/core#altLabel>

Example (text/turtle):

```
<https://www.purl.org/SomeOntologyClass> rdf:type owl:Class ;  
    skos:altLabel "intelligent grid"@en .
```

Alternative properties:

- <http://www.geneontology.org/formats/oboInOwl#hasExactSynonym>
- http://purl.obolibrary.org/obo/IAO_0000118

SHACL validation rules:

- `sh:datatype rdf:langString`

4.3 Optional metadata

4.3.1 Term editor

When an ontology is collaboratively edited, it may help to note down the editor(s) of a term. Questions about the term may be directed forwarded to them. It is also useful to credit the contributors of ontology development. We recommend to provide the contributors via their [ORCIDs](#).

Recommended property: <http://purl.org/dc/terms/contributor>

Example (text/turtle):

```
<https://www.purl.org/SomeOntologyClass> rdf:type owl:Class ;  
    dcterms:contributor <https://orcid.org/0000-0002-1584-4316> .
```

Alternative properties:

- <http://purl.org/dc/elements/1.1/contributor>
- <https://schema.org/contributor>

SHACL validation rules:

- `sh:nodeKind sh:IRI`

4.3.2 Date modified

A useful information for editors and users of a term is also when the term was last modified. This may be informative for editorial processes and update cycles but also to assess whether a term is still up to date.

Recommended property: <http://purl.org/dc/terms/modified>

Example (text/turtle):

```
<https://www.purl.org/SomeOntologyClass> rdf:type owl:Class ;  
    dcterms:modified "2025-04-15T10:13:00"^^xsd:dateTime .
```

Alternative properties: n/a

SHACL validation rules:

- `sh:xone ([sh:datatype xsd:date;] [sh:datatype xsd:dateTime ;] [sh:datatype xsd:dateTimeStamp;])`

4.3.3 Editor note

For the editing process, it may be helpful to add notes about the current editorial status of a term or tasks that still need to be performed. The editorial comment should be tagged for a language. There may be several editorial comments for a term. They do not necessarily need to be published with the ontology, but could be managed as an owl:import. We recommend to maintain them, so that editorial decisions remain accessible and traceable. The editorial notes could also carry a date and the editor name.

Recommended property: [editor note \(IAO:0000116\)](#)

Example (text/turtle):

```
<https://www.purl.org/SomeOntologyClass> rdf:type owl:Class ;
  obo:IAO_0000116 "[2024-04-14][SA] The class should be deprecated."@en .
```

Alternative properties:

- <http://www.w3.org/2004/02/skos/core#editorialNote>

SHACL validation rules:

- `sh:datatype rdf:langString`

4.3.4 Example of usage

In order to better understand how a term can be applied, an example of usage can be helpful. This can for example be provided in the form of a phrase describing how a term should be used, by an example instance or sub-entity of a term or by some example code.

Recommended property: [example of usage \(IAO:0000112\)](#)

Example (text/turtle):

```
<https://www.purl.org/SomeOntologyClass> rdf:type owl:Class ;
  obo:IAO_0000112 ""@en .
```

Alternative properties:

- <http://www.w3.org/2004/02/skos/core#example>
- <http://purl.org/vocab/vann/example>

SHACL validation rules:

- `sh:xone ([sh:datatype rdf:langString ;] [sh:nodeKind sh:IRI])`

4.3.5 Defined by

In some contexts it may be relevant to provide the original source of a term, e.g. if it is re-used in an ontology. In this case, the IRI of the original ontology should be provided. For maximum traceability, we recommend to not only use the ontology IRI to the latest version of an ontology, but a version IRI. This allows to understand which version of a term has been used.

Recommended property: <http://www.w3.org/2000/01/rdf-schema#isDefinedBy>

Example (text/turtle):

```
<https://www.purl.org/SomeOntologyClass> rdf:type owl:Class ;
  rdfs:isDefinedBy <https://www.purl.org/SomeOntology> .
```

Alternative properties: n/a

SHACL validation rules:

- `sh:nodeKind sh:IRI`

4.3.6 Issue link

Sometimes it makes sense to add the link to an issue to the annotations of a term. Especially if there is no consent about the term, the discussion may involve more viewpoints than a single editorial note is capable to encompass.

Recommended property: [term tracker item \(http://purl.obolibrary.org/obo/IAO_0000233\)](http://purl.obolibrary.org/obo/IAO_0000233)

Example (text/turtle):

```
<https://www.purl.org/SomeOntologyClass> rdf:type owl:Class ;
obo:IAO_0000233 <https://github.com/SomeOrganisation/SomeOntology/issues#12>
```

Alternative properties: n/a

SHACL validation rules:

- `sh:nodeKind sh:IRI`

4.3.7 Comments

If there is anything else to say about a term, this may fit into a general purpose comment.

Recommended property: <http://www.w3.org/2000/01/rdf-schema#comment>

Example (text/turtle):

```
<https://www.purl.org/SomeOntologyClass> rdf:type owl:Class ;
  rdfs:comment "Do not forget to colour some easter eggs!
<https://www.publicdomainpictures.net/pictures/160000/velka/oeufs-de-paques-1457782172BLx.jpg>"@en.
```

Alternative properties: n/a

SHACL validation rules:

- `sh:datatype rdf:langString`

4.4 Tabular overview - recommendations

section	Recommended property	Mandatory	Recommended	Optional	Cardinality
Preferred label	http://www.w3.org/2000/01/rdf-schema#label	x			1..* (only one per language)
Definition	http://www.w3.org/2004/02/skos/core#definition	x			1..* (only one per language)
Deprecation marker	http://www.w3.org/2002/07/owl#deprecated	x (if applicable)			0..1
Obsolescence reason	has obsolescence reason (IAO:0000231)	x (if applicable)			0..1
Term replacement	term replaced by (IAO:0100001)	x (if applicable)			0..*
Synonyms/ alternative labels	http://www.w3.org/2004/02/skos/core#altLabel		x		0..*
Term editors	http://purl.org/dc/terms/contributor			x	0..*
Date modified	http://purl.org/dc/terms/modified			x	0..1
Editor Note	editor note (IAO:0000116)			x	0..*

section	Recommended property	Mandatory	Recommended	Optional	Cardinality
Example of usage	example of usage (IAO:0000112)			x	0..*
Defined by	http://www.w3.org/2000/01/rdf-schema#isDefinedBy			x	1
Issue link	term tracker item (http://purl.obolibrary.org/obo/IAO_0000233)			x	0..*
Comments	http://www.w3.org/2000/01/rdf-schema#comment			x	0..*
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4.5 Relations to related work

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